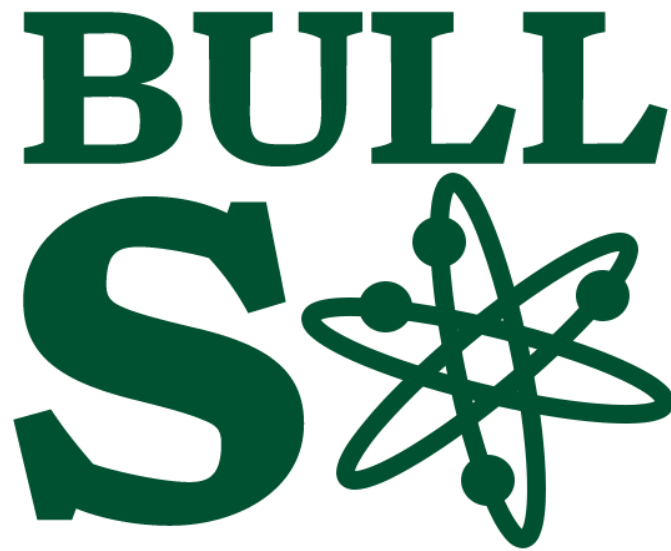


BullSO 2026 Invitational
Machines B



Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Number: _____

Team Name: _____

Participant Names: _____

Total Score: _____/30

Rank: _____

Part I: Multiple Choice (1 point each)

1. A lever has an effort arm of 2.0 m and a resistance arm of 0.5 m. What is the mechanical advantage?
 - a. 0.25
 - b. 1.0
 - c. 4.0
 - d. 8.0
2. Which of the following best describes an inclined plane?
 - a. A device that changes the direction of a force only
 - b. A flat surface set at an angle to reduce the force needed to lift an object
 - c. A rotating wheel with a groove
 - d. A rigid bar that pivots about a fulcrum
3. A pulley system has 6 supporting rope segments. What is the ideal mechanical advantage?
 - a. 2
 - b. 3
 - c. 6
 - d. 12
4. If a machine has an efficiency of 75% and an ideal mechanical advantage of 8, what is its actual mechanical advantage?
 - a. 6
 - b. 8
 - c. 10.7
 - d. 75
5. Which simple machine is most directly used in a screw?
 - a. Lever
 - b. Wedge
 - c. Inclined plane
 - d. Pulley
6. A 300 N load is lifted using a machine with an actual mechanical advantage of 5. How much effort force is required?
 - a. 60 N
 - b. 1500 N
 - c. 5 N
 - d. 300 N
7. Which factor most directly reduces the efficiency of real machines?
 - a. Mechanical advantage
 - b. Distance moved by the load
 - c. Friction

- d. Mass of the load
8. A first-class lever always has:
- a. The load between the effort and fulcrum
 - b. The effort between the load and fulcrum
 - c. The fulcrum between the effort and load
 - d. Equal arm lengths
9. The work input to a machine is 600 J and the work output is 420 J. What is the efficiency?
- a. 30%
 - b. 42%
 - c. 70%
 - d. 143%
10. Increasing the length of the ramp of an inclined plane while keeping the height constant will:
- a. Increase the work required
 - b. Decrease the work required
 - c. Increase the force required
 - d. Decrease the force required
11. Which class of levers always has an ideal mechanical advantage greater than 1?
- a. Class 1
 - b. Class 2
 - c. Class 3
 - d. Class 4
12. Archimedes famously once said:
- a. "Give me a place to stand, and a lever long enough, and I will move the world"
 - b. "Gravity works on an elaborate system of levers and pulleys"
 - c. "Every contraption, miniscule or gargantuan, can be broken down into the six simple machines"
 - d. "Six seven"

Part II: Free Response (3 points each)

Include units in your answer if applicable

- 13) A second-class lever is used to lift a 400 N load. The resistance arm is 0.40 m long and the effort arm is 1.20 m long.
- Calculate the ideal mechanical advantage.
 - Determine the effort force required (assume ideal conditions).
 - Explain one real-world factor that could increase the required effort force.
- 14) A block-and-tackle pulley system is used to lift a 600 N crate. The system has an efficiency of 80% and an ideal mechanical advantage of 6.
- Calculate the actual mechanical advantage.
 - Determine the effort force required.
 - Explain why the efficiency is less than 100%.
- 15) An object with a mass of 50 kg is pushed up a frictionless ramp that is 5.0 m long and 1.25 m high.
- Calculate the ideal mechanical advantage of the ramp.
 - Determine the force required to push the object up the ramp (use $g = 9.8 \text{ m/s}^2$).
 - Compare the work done lifting the object straight up versus using the ramp.
- 16) A machine requires an input force of 120 N applied over a distance of 4.0 m to lift a load 1.0 m vertically. The load weighs 360 N.
- Calculate the work input.
 - Calculate the work output.
 - Determine the efficiency of the machine.

- 17) You are designing a machine to lift heavy equipment in a laboratory.
- a) Identify two different simple machines you could combine.
 - b) Explain how combining these machines would reduce the required effort force.
 - c) Discuss one trade-off of increasing mechanical advantage in real machines.
- 18) A wooden plank is tilted to a maximum angle of 30 degrees to the floor before a block initially stationary on the plank starts to slide off the plank towards the floor. The block has a mass of 300 g. **Use $g = 10$**
- a) Calculate the static coefficient of friction (leave in trig) (EX: $\sin(20)$)
 - b) Calculate the maximum frictional force between the plank and the block
 - c) How would adding a heavier block of the same material change the angle at which it starts sliding off the plank?